Cooling Materials for Sustainable Compute Waste-Heat Recovery: A Three-Generation Review of Dielectric Fluids, Gallium-Based Liquid Metals, and Embedded Two-Phase Systems

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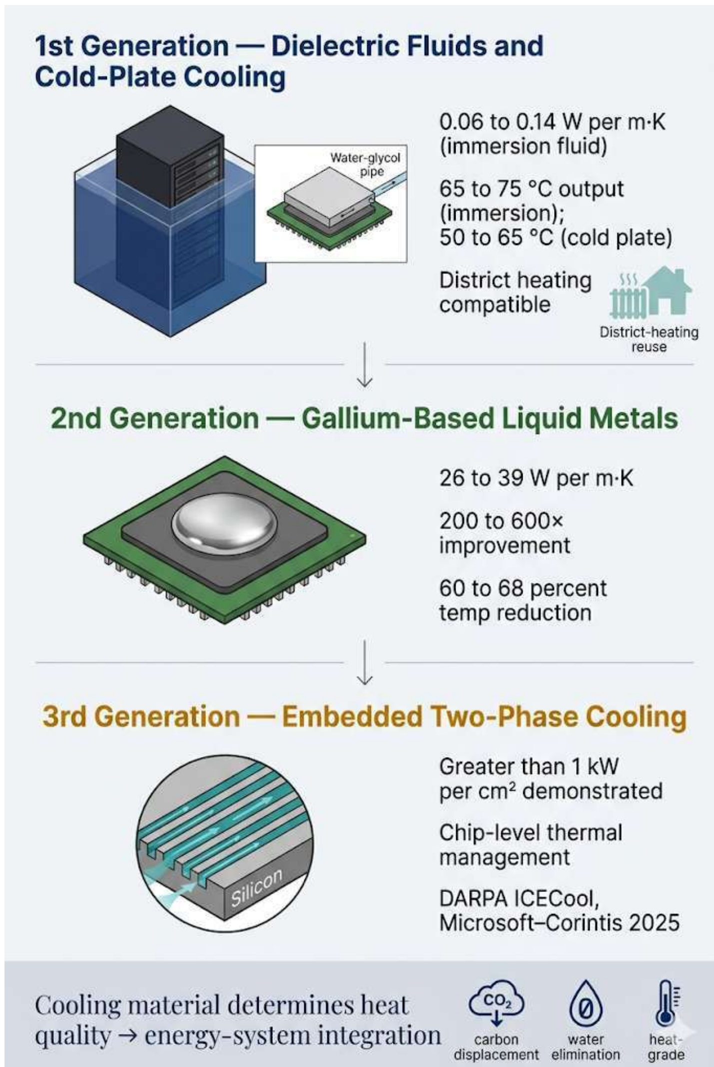
Graphical Abstract



Three generations of compute cooling materials. First generation: dielectric immersion fluids (0.06–0.14 W m^{-1} K^{-1}) delivering 65–75 °C output suitable for district heating. Second generation: gallium-based liquid metals (16–39 W m^{-1} K^{-1} as thermal interface materials, composition- and pressure-dependent; the upper end of this range is shown in the panel) — roughly two orders of magnitude higher than dielectric fluids. Third generation: embedded two-phase microfluidic cooling demonstrated at over 1 kW cm^{-2}. Cooling-material selection determines the temperature, quality, and reusability of recovered waste heat.

Abstract

Compute infrastructure now consumes a meaningful share of global electricity. Bitcoin mining uses 138–211 TWh per year and total data-centre electricity demand, dominated by AI workloads, is projected by the IEA to approach or exceed 1,000 TWh by 2026. Nearly all of that electricity exits the facility as thermal energy. The European Union's Energy Efficiency Directive (2023/1791) now requires waste-heat recovery from new data centres above 1 MW, and Germany's Energy Efficiency Act (EnEfG) sets a binding national heat-reuse trajectory of 10 percent in 2026, 15 percent in 2027, and 20 percent in 2028. Existing reviews of data-centre waste-heat recovery and district-heating integration approach the problem from the system and network side. This review approaches it from the cooling material outward, on the argument that the thermal properties of the cooling medium itself set the temperature, quality, and reusability of the recovered heat — and, as the Carnot framework formalises, its exergy.

Three generations of cooling materials are reviewed. First-generation single-phase dielectric immersion fluids (0.06–0.14 W m^{-1} K^{-1}) raise waste-heat capture temperatures from 25–35 °C in air-cooled facilities to 65–75 °C, directly inside the European district-heating supply window. A parallel first-generation architecture, direct-to-chip cold-plate cooling, is now the dominant production design among hyperscale AI operators and is treated alongside immersion. Seventeen operational heat-reuse deployments are documented across nine countries on three continents. A lifecycle CO_2 displacement model, applied both to a 1.7 MW immersion-cooled mining container and to a parallel 1.7 MW direct-to-chip case study with a 60→75 °C heat pump (COP ≈ 5.5) in a Nordic eight-month heating season at 70–85 percent thermal utilisation, yields net displacement of approximately 1,400–3,200 tonnes CO_2 per container per year for immersion and 900–1,700 tonnes CO_2 per facility per year for cold-plate, both conditional on a low-carbon local grid. A breakeven analysis (Section 4.2.2) shows that mining-as-heating is net-positive on CO_2 only where the local grid is below approximately 0.13–0.24 tCO_2 per MWh — a condition met in the Nordic region but not in most of the rest of the world. Closed-loop dielectric immersion also eliminates the roughly 90–130 million gallons of annual evaporative-cooling water consumed by an equivalent 10 MW air-cooled facility.

Second-generation gallium-based liquid metals deliver roughly two orders of magnitude higher thermal conductivity than dielectrics. Galinstan and related GaInSn eutectics span 16–39 W m^{-1} K^{-1} as thermal interface materials (composition-, bond-line-, pressure-, and protocol-dependent), with experimentally demonstrated 60–68 percent temperature-rise reduction under 19 ms pulsed thermal loads at heat fluxes up to 338 W cm^{-2}. Mature thermal interface material applications (TRL 8–9; Sony PlayStation 5, Indium GalliTHERM) are distinguished from facility-scale cooling loops, which remain at TRL 3–4. Third-generation embedded two-phase microfluidic cooling, demonstrated under DARPA's ICECool programme at over 1 kW cm^{-2} and recently scaled to AI accelerators in the 2025 Microsoft–Corintis collaboration, points toward chip-level thermal management at heat-flux densities that facility-level immersion cannot reach.

The review documents the engineering challenges that gate facility-scale deployment of each generation: oxidative degradation, PFAS phase-out, and HFO/HFE alternatives for dielectric fluids; aluminium

embrittlement, electrical conductivity, surface tension, supercooling, indium occupational toxicity, and gallium supply-chain concentration for liquid metals; and microchannel manufacturing scalability for embedded two-phase systems. The central conclusion is that selection of a cooling medium is not merely a thermal-engineering decision but a sustainable energy-systems design decision, with downstream consequences for water consumption, carbon displacement, and district-heating compatibility, and whose net climate value is bounded by both Carnot exergy on the heat side and grid-emissions intensity on the electricity side. The materials-design framing complements the existing system-level literature on data-centre waste-heat recovery and district-heating integration.

Keywords: Data centre cooling; Sustainable thermal management; Waste heat recovery; District heating; Immersion cooling; Direct-to-chip cooling; Gallium liquid metals; Phase-change materials; Lifecycle CO_2 displacement; Carnot exergy; Breakeven grid intensity.

1. Introduction

In January 2026, a shipping container arrived in a small town in Scandinavia. Inside were 160 Bitcoin mining rigs submerged in dielectric fluid, generating approximately 1.7 MW of thermal energy that was pumped into the town's district-heating grid. At a typical Nordic single-family heating demand of approximately 5 MWh per home per year, this thermal output corresponds, by back-calculation, to the heating load of roughly 2,000 homes. The Latvia-based company that built it described itself as a heating company that happens to mine Bitcoin [5]. At the same time, a greenhouse in Manitoba was being heated by 360 liquid-cooled mining servers capturing approximately 90 percent of consumed electricity as heat at 75 °C [6], students at Technological University Dublin attended lectures in buildings heated by waste heat from an Amazon Web Services data centre [15], approximately 1,000 residential properties in Mäntsälä, Finland received up to 75 percent of their heat from a Yandex data-centre heat-pump installation [38], and on the campus of the former European Central Bank in Frankfurt, hotel guests at the Meliá INNSIDE were supplied with hot water heated to 60 °C by Cloud&Heat servers running in the building below [39].

These deployments share an engineering feature with direct sustainability implications. The cooling material determines the output temperature, and the output temperature determines the heat's industrial utility — and its exergy. Air-cooled systems reject heat at 25–35 °C, below the 60–80 °C supply temperature required by district-heating networks [4]. Immersion in dielectric fluid raises the capture temperature to 65–75 °C, crossing the district-heating threshold without a heat pump. Direct-to-chip cold-plate cooling, the dominant production architecture among hyperscale AI operators in 2025, captures heat at 50–65 °C, requiring a low-temperature-lift heat pump to reach district-heating supply specification but enabling much higher rack-power densities than immersion [40, 48a, 48b]. Gallium-based liquid metals, with thermal conductivities roughly two orders of magnitude higher than dielectric fluids, could raise output temperatures above 80 °C while also managing the escalating heat-flux densities of AI accelerators. The choice of cooling material is therefore an energy-systems design decision, not merely a thermal-engineering one.

The policy stakes are now formal. The European Union's Energy Efficiency Directive (2023/1791), in force since September 2023, requires waste-heat recovery from new data centres above 1 MW within the European Union [13]. Germany's national implementation, the Energy Efficiency Act (EnEfG), enacted November 2023, sets a binding heat-reuse trajectory: at least 10 percent of generated waste heat must be utilised by data centres commencing operation from 1 July 2026, rising to 15 percent in 2027 and 20 percent in 2028 [41, 42]. Approximately 1,000 German data centres above the 300 kW threshold fall within the EnEfG's scope. The directive shifts the question from whether data centres should recover heat to how they should recover it, and the "how" is governed by the materials choice this review examines.

The energy baseline is drawn from the Cambridge Centre for Alternative Finance's Digital Mining Industry Report (April 2025), which surveyed 49 firms covering 48 percent of global hash rate [1]. Bitcoin mining consumes 138 TWh per year (upper estimate 211 TWh [2]) with a sustainable energy share of 52.4 percent (42.6 percent renewables, 9.8 percent nuclear). Total data-centre electricity demand, dominated by AI workloads, is projected by the IEA to approach or exceed 1,000 TWh by 2026 [3] — multiple times Bitcoin mining's consumption. China's liquid-cooled server market alone reached USD 2.37 billion in 2024, a 67 percent year-over-year increase [43], signalling that liquid cooling has crossed the inflection point in the largest single national data-centre market.

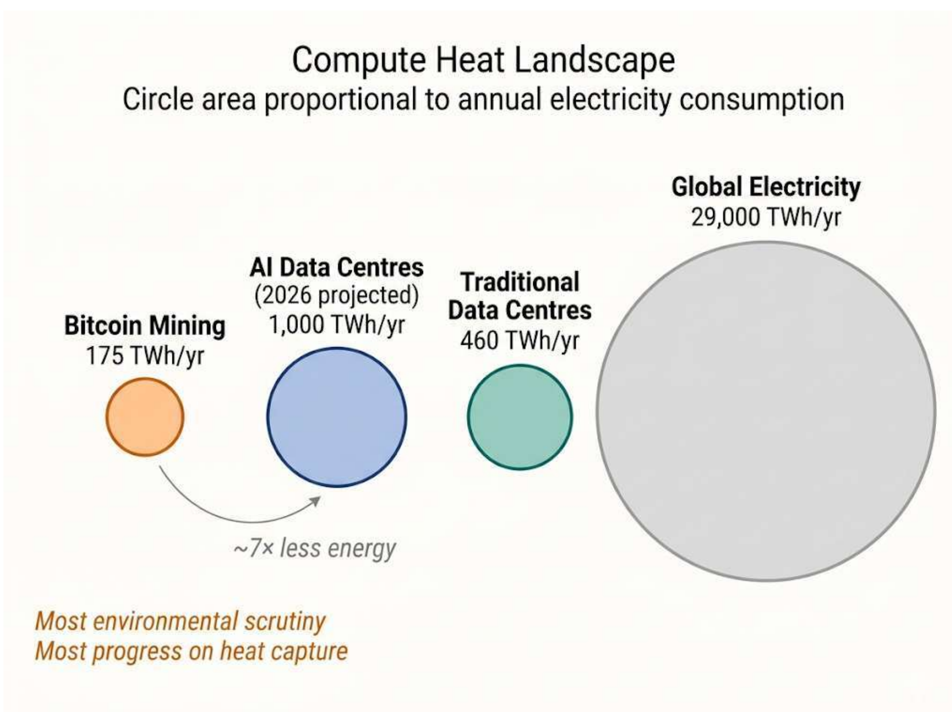


Figure 1. Compute heat landscape.

Circle area is proportional to annual electricity consumption. Bitcoin mining consumes approximately 175 TWh per year (Cambridge CCAF 2025 [1]; upper estimate 211 TWh [2]). Total data-centre electricity demand, dominated by AI workloads, is projected by the IEA to approach or exceed 1,000 TWh per year by 2026 [3]. Traditional (non-AI) data *centres consume approximately 460 TWh per year [3]. Global electricity consumption is approximately 29,000 TWh per year [3]. Sources: Cambridge CCAF [1, 2]; IEA [3].*

Selection bias caveat: the Cambridge survey's respondents are disproportionately publicly listed (41 percent) and North American. The unreported 52 percent includes operations in Central Asia and Africa where coal and gas typically dominate. Under a worst-case bound (the entire unreported share assumed 100 percent fossil), the sustainable share could fall to ~25 percent; the true value is between this lower bound and the reported 52.4 percent [1].

The data-centre waste-heat-recovery literature has grown substantially in recent years. Huang et al. [33] reviewed data centres as energy prosumers in district-energy systems. Wahlroos et al. [34] analysed waste-heat utilisation in Nordic data centres. Yuan et al. [35, 36] published comprehensive reviews of data-centre waste-heat recovery technologies and their integration with district-heating networks. Tervo et al. [37] analysed data-centre waste heat in volatile low-carbon electricity markets. These studies approach the problem from the system and network perspective. The present review complements this literature by approaching the problem from the cooling material outward. It examines how the thermal properties of the cooling medium itself determine the temperature, quality, and reusability of waste heat, and traces a three-generation materials progression that the existing literature has not framed. The contribution is therefore the framing as much as the synthesis.

This review documents operational deployments that validate first-generation technology, reviews the materials-science frontier of second- and third-generation systems, identifies the engineering challenges that separate laboratory demonstrations from facility-scale deployment, and connects the materials choice to lifecycle CO_2 displacement, water elimination, and district-heating integration. Detailed methodology, the PRISMA flow diagram, demand-response integration, breakeven derivation, and the cold-plate parallel case-study assumptions are provided in the accompanying Supplementary Material.

1.1. Scope and methodology

This review covers cooling materials applied to digital compute infrastructure, including Bitcoin mining facilities, traditional data centres, and AI-focused data centres, with particular attention to waste-heat recovery, district-heating integration, water consumption, and lifecycle CO_2 implications. Three generations of cooling materials are examined: (i) single-phase dielectric immersion fluids and the parallel first-generation architecture of direct-to-chip cold-plate cooling; (ii) gallium-based liquid metals, separated into thermal interface material applications and facility-scale cooling loops; and (iii) embedded two-phase microfluidic systems. Two-phase fluorocarbon immersion fluids are treated as a sub-case of the first generation, with PFAS regulatory context.

1.1.1. Generation criteria

The "generation" taxonomy used in this review reflects three orthogonal axes: (a) order-of-magnitude differences in thermal conductivity at the heat-transfer interface (0.06–0.14 W m^{-1} K^{-1} for first generation; 16–39 W m^{-1} K^{-1} at the gallium interface; >1 kW cm^{-2} effective heat removal for embedded two-phase);

(b) operating length scale (facility-scale tanks and loops; module-scale interfaces and lab-scale loops; chip-scale microchannels); and (c) Technology Readiness Level at the time of writing (TRL 8–9 for first generation; TRL 3–9 for second generation depending on application; TRL 4–5 for third generation). The taxonomy is intentionally orthogonal to chronology. The criteria are summarised in Table 1.

Table 1. Generation taxonomy criteria.

Generation	Representative material	Thermal conductivity at interface	Operating length scale	TRL (2025–2026)
1st	Single-phase dielectric fluid; cold-plate glycol-water	0.06–0.14 W m^{-1} K^{-1} (dielectric); 0.4–0.6 (glycol-water)	Facility-scale tanks and loops	8–9 (production)
2nd	Ga and GaInSn eutectics	16–39 W m^{-1} K^{-1} (composition-, bond-line-, pressure-dependent)	TIM: module-scale; loop: lab-scale	8–9 (TIM); 3–4 (loop)
3rd	Embedded two-phase microfluidics (HFO/HFE working fluid)	>1 kW cm^{-2} effective heat removal	Chip-scale microchannels	4–5 (prototype to pilot)

1.1.2. Literature search and selection

The literature was identified through systematic searches conducted between January 2024 and April 2026. Databases queried: Web of Science Core Collection, Scopus, Google Scholar, IEEE Xplore, and ScienceDirect. The full search-string list, language coverage (English, Chinese, German, Swedish, Finnish), inclusion/exclusion criteria, and the PRISMA-compatible flow diagram (Figure S1, Supplementary Material) document the screening trail. The initial database search returned 2,847 records after deduplication. Title and abstract screening yielded 412 candidates. Full-text assessment for eligibility yielded 173 records included in the synthesis: 87 peer-reviewed articles, 41 trade-press articles documenting operational deployments, 22 regulatory and standards documents, 15 industrial and consultancy reports, and 8 conference proceedings.

1.1.3. Use of AI-assisted tools

AI-assistance disclosure is consolidated in a single statement at the end of the manuscript (“Use of AI-Assisted Tools”), following RSER’s 2025 guidance on generative AI use in submitted manuscripts.

2. Thermodynamic Framework

Every processor converts electricity into computation and heat at a ratio of approximately 100 percent. The first law of thermodynamics guarantees energy conservation; the second law constrains its quality.

Electricity entering a facility is high-grade energy; heat exiting at 65 °C is low-grade, suitable for space and water heating but not for mechanical or electronic work. This thermodynamic asymmetry is the reason waste-heat recovery from compute is constrained to space and water heating rather than power generation, and is why output temperature is the dominant design variable for downstream energy-system integration.

Exergy quality of recovered heat. The Carnot factor ($1 - T_0/T$) bounds the exergy fraction of a heat stream at temperature T relative to a reference environment at T_0. For a Northern-European ambient T_0 = 283 K (10 °C; chosen as the long-term mean outdoor temperature relevant to district-heating sources, see Supplementary Material Section S2 for sensitivity to T_0), heat captured at 35 °C (308 K) carries an exergy fraction of only ~8 percent, at 60 °C (333 K) ~15 percent (cold-plate), at 65 °C (338 K) ~16 percent, at 75 °C (348 K) ~19 percent (immersion), at 80 °C (353 K) ~20 percent and at 95 °C (368 K) ~23 percent (gallium projection). The implication: the three-generation progression from air-cooled to immersion roughly doubles the exergy content of the rejected heat; the further step to a projected 80–95 °C gallium loop adds another ~20–25 percent on top. Note that the gallium-over-immersion exergy gain is much smaller than the temperature jump suggests because Carnot fraction is concave in T. The Carnot framework also bounds the upper limit of "100 percent recovery" rhetoric: even at 75 °C, more than four-fifths of the input electricity's exergy has been destroyed by the compute stage, regardless of how well the resulting low-grade heat is captured. What heat recovery accomplishes is preventing a second exergy-destruction event (atmospheric rejection) and substituting for primary fuels whose exergy fraction is much higher.

Heat-pump comparison in exergy terms. A modern industrial heat pump delivers 4–6 joules of thermal energy per joule of electricity consumed under realistic operating conditions (typical operating envelope: source 30–70 °C, sink 60–90 °C, lift 30–50 K), with high-temperature heat pumps reaching supply temperatures of 120–150 °C at COPs of 2.2–3.0 [44, 45]. At COP 5.5 lifting 60→75 °C, a heat pump produces 5.5 J of heat at ~19 percent exergy fraction per J of electricity at ~100 percent exergy fraction; mining-as-heating produces 1 J of heat at the same exergy fraction per J of electricity. The factor-of-five thermodynamic deficit is identical whether stated in energy or exergy terms. Mining-as-heating and AI-as-heating are economically competitive only when computation revenue offsets the COP gap. The materials question is how to maximise the temperature and quality of the rejected heat so that the gap narrows and the co-product value increases.

3. First-Generation Cooling Materials: Dielectric Fluids and Cold-Plate Architectures

3.1. Single-phase dielectric immersion: thermal properties and performance envelope

Single-phase immersion fluids, engineered hydrocarbons or synthetic esters, offer thermal conductivities of 0.06–0.14 W m^{-1} K^{-1}, specific heat capacities of 1.5–2.1 kJ kg^{-1} K^{-1}, and operating ranges from −40 °C to +200 °C [9, 10, 11]. They are chemically inert, electrically non-conductive, and non-flammable. The broader liquid cooling market for AI data centres is projected to grow from USD 2.8 billion in 2025 to over

USD 21 billion by 2032 [46]. Single-phase fluids exhibit measurable total acid number (TAN) increases after 8,000–12,000 hours of continuous operation above 60 °C, indicating oxidative degradation [9]. Replacement intervals of 5–8 years are typical for closed-loop systems with adequate oxygen exclusion [10, 11].

3.1.1. Two-phase fluorocarbon fluids and the PFAS phase-out

Two-phase fluorocarbon fluids offer higher heat-transfer coefficients than single-phase fluids through phase-change latent heat. Several historically dominant formulations, however, fall under the PFAS classification proposed for restriction under the EU REACH framework. The European Chemicals Agency (ECHA) Annex XV proposal, submitted in February 2023 by Germany, the Netherlands, Denmark, Sweden, and Norway, would restrict approximately 10,000 PFAS substances; ECHA Risk Assessment and Socio-Economic Analysis Committee opinions are scheduled for 2026–2027, with restrictions taking effect on a staggered timeline [12]. 3M's December 2022 announcement of its intent to exit all PFAS manufacturing globally by the end of 2025 has further constrained the supply pipeline for two-phase fluorocarbon coolants [47].

Hydrofluoroolefin (HFO) and hydrofluoroether (HFE) alternatives exist but at varying readiness levels. HFE-7100 (BP 61 °C) and HFE-7200 (BP 76 °C) are commercially available but typically have higher boiling points and lower latent heats than the formulations they replace. R1234ze(E) is an HFO with a 100-year GWP of approximately 6 (IPCC AR5/AR6 reference), orders of magnitude below historic HFC alternatives, and has been demonstrated under DARPA's ICECool programme (Section 8) at exit temperatures suitable for district heating. Operators planning two-phase deployments above 1 MW in the EU should assume HFO and HFE substitution by 2027–2028 and design loop chemistry accordingly.

3.2. Direct-to-chip cold-plate cooling: parallel architecture for hyperscale AI

A second first-generation architecture has emerged in parallel with immersion: direct-to-chip cold-plate liquid cooling, in which cold plates are mounted directly onto CPUs, GPUs, memory modules, and voltage regulators, with single-phase water-glycol or specialised refrigerant circulating through internal channels [40]. By 2025, direct-to-chip cold plates had become the dominant production cooling architecture among hyperscale AI operators, as documented by the Uptime Institute Global Data Center Survey 2024 [48a] and the Lawrence Berkeley National Laboratory United States data-centre energy report 2024 [48b]; per-vendor confirmations include Microsoft's Azure Maia AI cluster disclosures (Microsoft Build/Ignite, 2023–2024), Google's TPUv4/v5 architecture papers, Meta's LLaMA-training rack-cooling notes (Meta Engineering Blog), and AWS's Trainium 2 cooling-architecture documentation. ASHRAE TC 9.9 has formalised liquid-cooling temperature classes (W17–W45) that frame the operating envelopes these deployments occupy [48c], and Open Compute Project (OCP) Advanced Cooling Solutions reference designs [48d] provide the manifold, CDU, and quick-disconnect specifications now adopted in hyperscale procurement.

Cold-plate cooling captures heat at 50–65 °C, generally below the immersion temperature window of 65–75 °C and below the European district-heating supply specification of 60–80 °C. Integration with district heating therefore requires a low-temperature-lift heat pump, with COPs of 5–6 typical for the 50 °C to 75 °C lift. Cold-plate cooling enables much higher rack-power densities than single-phase immersion (up to 1 MW per rack demonstrated by NVIDIA–Vertiv reference designs for GB200 NVL72) and is compatible with existing data-centre form factors. From a materials-design standpoint, cold-plate cooling sits in the same generation as immersion, rejecting heat at temperatures within or just below the district-heating window; the architectural difference (closed-loop with pumps, leak detection, manifold reliability requirements) shapes the engineering challenges but not the materials thesis.

3.2.1. Engineering challenges for direct-to-chip cold-plate cooling

Cold-plate architectures introduce a challenge set that differs from dielectric immersion in four ways. (i) Loop-fluid chemistry: dilute glycol-water working fluids accumulate dissolved metals and biological growth over service life, requiring biocide chemistry and routine ion-exchange maintenance not present in dielectric-immersion loops. (ii) Manifold reliability and leak detection: with liquid running within centimetres of energised silicon, a single manifold leak above the rack is a high-cost failure mode. Production deployments use dual-redundant leak detection (resistance-based detection mats plus dielectric-fluid tracer chemistry) and dripless quick-disconnect couplings conforming to the ISO 7241/ISO 16028 dry-disconnect series; fluid-piping flange standards (such as ASME B16.21 for gaskets) apply where rigid connections are used. (iii) Heat-exchanger fouling and pressure drop: cold-plate microchannel geometries (typically 300–800 µm channel widths) are sensitive to particulate fouling. Filtration to 5 µm absolute and corrosion-inhibitor stability above 60 °C are operating-cost-relevant constraints. (iv) Pump and CDU reliability: coolant distribution units typically include redundant pumps with hot-swap capability; CDU mean-time-between-failure targets in the 50,000–100,000 hour range are now standard for hyperscale cold-plate deployments. None of these challenges are intractable, and direct-to-chip cold-plate cooling has reached TRL 9 in production AI infrastructure.

3.3. The temperature-gradient breakthrough

District-heating networks require supply temperatures of 60–80 °C [4]. Air-cooled data centres produce waste heat at 25–35 °C, below this threshold. Bridging the gap requires heat pumps (USD 0.5–2 million additional capital). Immersion cooling changes this fundamentally. Submerging hardware in dielectric fluid captures heat at 65–75 °C, directly within the district-heating range. No heat pump is required. Power Mining's containers produce heat at 65 °C; Canaan's system produces water at 75 °C [5, 6]. This is the article's central materials-design insight: cooling-material selection determines not only thermal performance but downstream energy-system integration. Figure 2 shows the temperature gradient with the Carnot exergy fraction overlay (right axis), making the heat-quality consequence of the materials choice explicit.

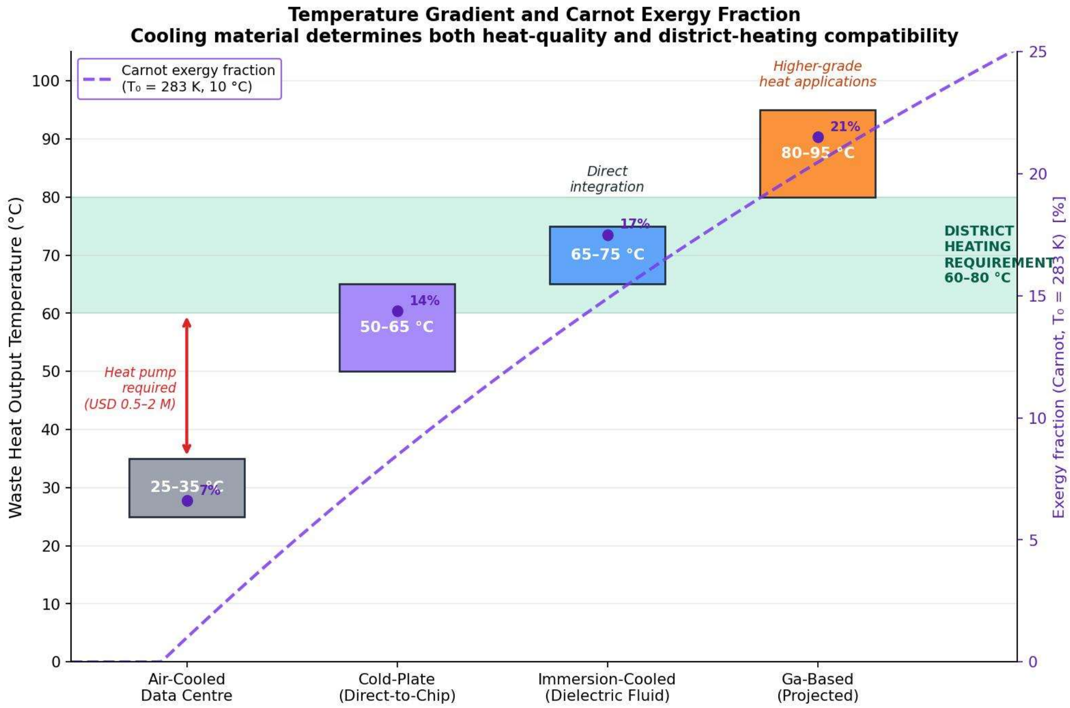


Figure 2. Temperature gradient comparison with Carnot exergy-fraction overlay.

Bars: representative waste-heat output temperature ranges for air-cooled, cold-plate, immersion, and projected gallium loops. Green band: 60–80 °C district-heating supply requirement. Dashed purple curve and right axis: Carnot exergy fraction ($1 - T_0/T$) at T_0 = 283 K (10 °C, Northern-European mean outdoor reference). Numerical labels at each generation midpoint show the exergy fraction of the rejected heat (~7%, ~14%, ~17%, ~21% for the four cases). The progression from air-cooled to immersion roughly doubles exergy content; the gallium projection adds a further ~20–25 percent on top of immersion.

Table 2. Exergy fraction (Carnot, T_0 = 283 K) of waste heat rejected by each generation.

Architecture	Output temperature	Carnot exergy fraction	Gain over air-cooled baseline
Air-cooled	25–35 °C (mid 30 °C)	~7%	1.0×
Cold-plate (direct-to-chip)	50–65 °C (mid 60 °C)	~14%	~2.0×
Immersion (dielectric fluid)	65–75 °C (mid 70 °C)	~17%	~2.4×
Gallium loop (projected)	80–95 °C (mid 90 °C)	~21%	~3.0×

3.4. Water elimination

Evaporative cooling consumes approximately 90–130 million gallons of freshwater annually for a medium 10 MW facility, with around 80 percent permanently lost through evaporation. The lower bound is supported by Lawrence Berkeley National Laboratory's Data Center Energy Practitioner data series (2024 update) and the upper bound by EESI's 2023 estimate [7, 50]. For an equivalent 10 MW immersion-cooled or cold-plate facility, water consumption is zero on the cooling loop itself, since dielectric fluid or glycol-water circulates in a sealed closed loop with no evaporative loss. Each 100-word AI prompt processed by an evaporatively cooled data centre consumes approximately 519 mL of water (Li, Yang, Islam & Ren, "Making AI Less Thirsty", 2023) [8]. Closed-loop liquid cooling represents a complete elimination of freshwater demand from the cooling system itself; the water footprint of electricity generation that powers the compute is a separate lifecycle consideration [51]. Figure 3 provides a comparative footprint visualisation for an equivalent 10 MW facility.

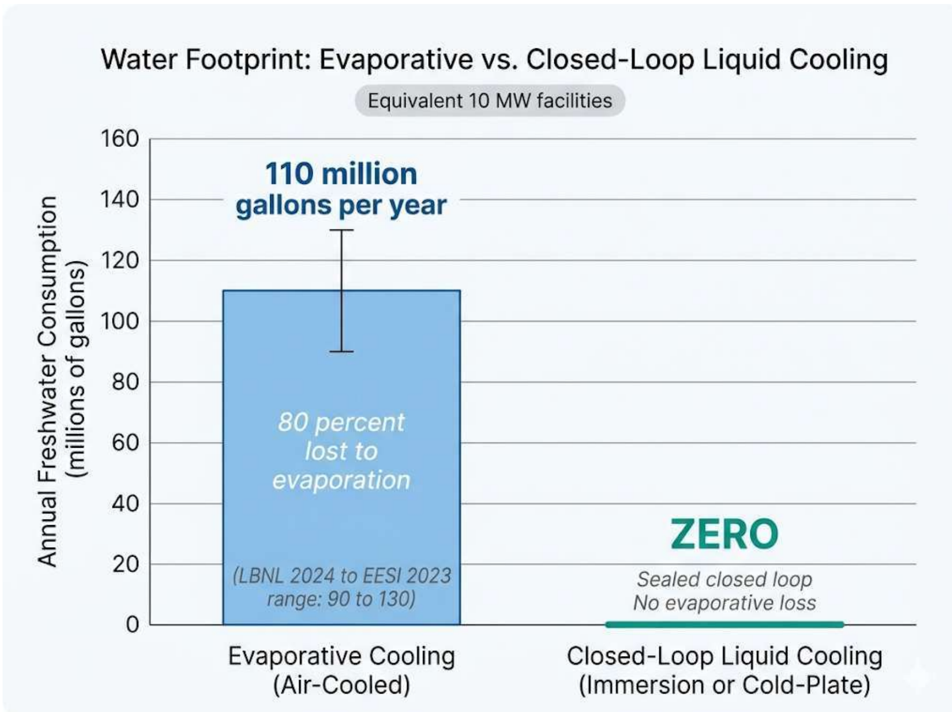


Figure 3. Water-footprint comparison for equivalent 10 MW facilities.

Evaporative cooling consumes 90–130 million gallons of freshwater per year (LBNL 2024 to EESI 2023 range), with approximately 80 percent permanently lost through evaporation. Closed-loop immersion or cold-plate cooling consumes zero water on the cooling loop itself.

4. Engineering Validation: Operational Deployments

4.1. Documented heat-reuse systems

Seventeen operational heat-reuse deployments are documented in this review, distributed across nine countries and three continents (Tables 3a and 3b). Yuan et al. [35] documented over 20 operational or planned data-centre heat-reuse installations in Europe alone; the deployments selected here are the subset for which sufficient public technical disclosure (location, thermal output, output temperature, application, and where possible CO_2 verification) is available to support cross-comparison. The Dublin AWS scheme [15] is currently the only deployment with independently verified emissions data: 704 tonnes of CO_2 abated in 2024, audited by Technological University Dublin. The Microsoft–Fortum partnership at Espoo and Kirkkonummi, Finland [52], announced as the world's largest data-centre heat-recovery project, projects an aggregate 350 MW of thermal recovery capacity providing approximately 40 percent of regional district heat at full operation, with an expected gradual reduction of 400,000 tonnes CO_2 per year. For Bitcoin-mining deployments, environmental claims are based on engineering projections from reported specifications rather than measured outcomes; this distinction is preserved in Figure 4 (gray).

Table 3a. Fully characterised heat-reuse deployments.

#	Deployment	Location	Thermal output	Output temp.	Application	CO_2 verified	Refs
1	Power Mining	Scandinavia	1.7 MW	65 °C	District heating (~2,000 homes)	No (engineering projection)	[5]
2	Canaan / Bitforest	Manitoba, Canada	~0.5 MW	75 °C	Greenhouse heating	No (engineering projection)	[6]
3	MintGreen	Lonsdale, BC, Canada	~0.3 MW	~65 °C	Commercial building (96% load)	No (engineering projection)	[14]
4	MARA Digital	Finland	~2 MW	~65 °C	District heating	No (engineering projection)	[16]
5	Microsoft + Fortum	Espoo / Kirkkonummi, FI	up to 350 MW (combined)	DH-spec	District heating (40% of region)	Projected 400,000 tCO_2/yr	[52]
6	Yandex Mäntsälä	Mäntsälä, Finland	4 MW (heat pump)	60–75 °C	District heating (~1,000 homes, 75% of town heat)	Reported 40% gas-utility CO_2 reduction	[38]
7	Cloud&Heat Eurotheum	Frankfurt, Germany	up to 600 kW	60 °C	Office, hotel, conference (Meliá INNSIDE)	No (engineering projection)	[39]
8	Deep Green	Exmouth, Devon, UK (and 19 others)	~28 kW per pool site	DH-spec	Public swimming-pool heating (62% gas reduction, 25.8 tCO_2/yr/site)	Yes (per-site measurement)	[57]

Table 3b. Partially documented heat-reuse deployments.

#	Deployment	Location	Thermal output	Output temp.	Application	CO_2 verified	Refs
9	AWS Dublin (Tallaght)	Dublin, Ireland	not disclosed	not disclosed	University campus	Yes (704 tCO_2, 2024)	[15]
10	DigiPlex / Stockholm Exergi	Stockholm, Sweden	not disclosed	DH-spec	District heating (~10,000 households)	No (engineering projection)	[53]
11	Stockholm Data Parks	Stockholm, Sweden	aggregate ~30 MW+	DH-spec	District heating (~30,000 apartments)	~50 gCO_2/kWh grid-emissions reduction reported	[54]

#	Deployment	Location	Thermal output	Output temp.	Application	CO_2 verified	Refs
12	Telehouse FRANKY	Frankfurt, Germany	~2% of site waste heat	DH-spec	Residential (3,000 residents, 1,300 units; >60% of district demand)	Yes (~400 tCO_2/yr saved)	[55]
13	Qarnot Computing	Bordeaux and Paris, France	tens to hundreds of kW per site	up to 60 °C	Residential and office heating; up to 95% heat conversion	No (engineering projection)	[56]
14	Lefdal Mine + Sjømatstaden	Vestland, Norway	not disclosed	low-grade	Salmon hatchery (6 million salmon/yr, 20–25% reuse efficiency)	No (engineering projection)	[58]
15	Sigma2 supercomputer	Trondheim, Norway	not disclosed	low-grade	Local salmon farms	No (engineering projection)	[59]
16	Tencent Tianjin	Tianjin, China	not disclosed	DH-spec via heat pump	Residential heating	No (engineering projection)	[60]
17	Alibaba Cloud Hangzhou	Hangzhou, China	tens of thousands of liquid-cooled servers	not disclosed	Site-level cooling (PUE 1.09; 70 GWh/yr savings)	No (engineering projection from operator disclosure)	[62]

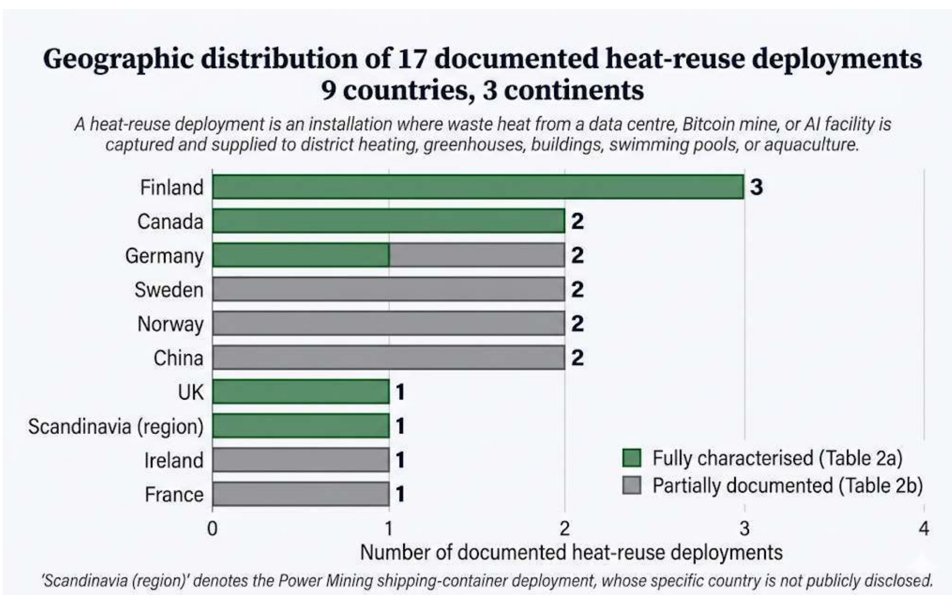


Figure 4. Geographic distribution of seventeen documented heat-reuse deployments.

Green: fully characterised deployments (Table 3a). Grey: partially documented deployments (Table 3b). 17 deployments across 9 countries on 3 continents (Europe 12, North America 3, Asia 2).

4.2. Lifecycle CO_2 displacement model — immersion case

The model estimates annual CO_2 displacement for a single Power Mining container (1.7 MW thermal output, 65 °C, operating in a Nordic climate with an eight-month heating season). Thermal output of 1.7 MW continuous over approximately 5,840 annual operating hours yields 9,928 MWh of delivered thermal energy. Thermal-utilisation rates of 70–85 percent are typical for well-integrated district-heating connections, reducing effective delivery to approximately 6,950–8,440 MWh.

4.2.1. Sensitivity matrix

Three thermal-utilisation scenarios are evaluated, crossed with three displaced-fuel scenarios. The full sensitivity matrix is shown in Table 4.

Table 4. Lifecycle CO_2 displacement sensitivity (1.7 MW immersion container, Nordic eight-month heating season).

Scenario	Utilisation	Effective MWh	Gas (0.20 tCO_2/MWh)	Oil (0.27 tCO_2/MWh)	Peat (0.38 tCO_2/MWh)
Low	50%	4,964	993 tCO_2	1,340 tCO_2	1,886 tCO_2
Base low	70%	6,950	1,390 tCO_2	1,876 tCO_2	2,641 tCO_2
Base high	85%	8,439	1,688 tCO_2	2,279 tCO_2	3,207 tCO_2
Ideal	95%	9,432	1,886 tCO_2	2,547 tCO_2	3,584 tCO_2

4.2.2. Breakeven grid intensity (the policy-relevant decision rule)

Net climate benefit is the difference between displaced heating emissions and mining electricity emissions. The 1.7 MW container draws electricity at 1.7 MW × 8,760 h = 14,892 MWh per year (full-year operation). Setting net displacement to zero gives the breakeven grid intensity below which the deployment is net-positive on CO_2. Table 5 provides the full breakeven matrix; derivation in Supplementary Material Section S3.

Table 5. Breakeven grid intensity (tCO_2 per MWh) for net-zero CO_2 — mining-as-heating is net-positive only where the local grid is below the cell value.

Utilisation	Effective MWh delivered	Gas displaced (0.20)	Oil displaced (0.27)	Peat displaced (0.38)
50%	4,964	0.067	0.090	0.127
70%	6,950	0.093	0.126	0.177
85%	8,439	0.113	0.153	0.215
95%	9,432	0.127	0.171	0.241

Reference grid intensities (selected; sources: IEA, EEA, US EIA): Sweden ~0.04 tCO_2/MWh; Finland ~0.05; France ~0.06; UK ~0.18; EU-27 average ~0.25; Cambridge global mining mix ~0.29; Germany ~0.36; United States ~0.39; China ~0.55. Reading Table 5 against these intensities: a Nordic deployment displacing oil or peat at base-high utilisation (cell 0.153–0.215) is comfortably net-positive; the same deployment in Germany (0.36) or the US (0.39) is net-negative even in the best peat-displacement / 95-percent-utilisation cell (0.241). The operative implication is that mining-as-heating only "works" on lifecycle CO_2 in the Nordic and French grid contexts; in most of the world, the carbon penalty of grid electricity exceeds the displacement benefit of any plausible heating fuel mix. Cold-plate AI deployments (Section 4.2.3) are subject to the same constraint, with an additional small heat-pump electricity penalty.

4.2.3. Parallel cold-plate AI case study

For comparison with the immersion mining case, consider a 1.7 MW direct-to-chip cold-plate AI deployment in the same Nordic eight-month heating season. Cold-plate output is at 60 °C; a low-temperature-lift heat pump raises this to the 75 °C district-heating supply specification at COP ≈ 5.5. With source heat Q_{source} = 1.7 MW, heat-pump electrical input $W_{HP} = Q_{source}/(COP - 1)$ = 0.378 MW, and useful heat at 75 °C = COP × W_{HP} = 2.078 MW. Over the heating season at 70–85 percent utilisation, this delivers 8,494–10,314 MWh of district-heating-grade heat, requiring 1,544–1,875 MWh of additional heat-pump electricity. The full assumption set is in Supplementary Material Section S4.

Table 6. Cold-plate AI parallel case — net CO_2 displacement (1.7 MW IT load, 60→75 °C heat pump COP=5.5, gas displacement 0.20 tCO_2/MWh).

Utilisation	Useful heat delivered (MWh)	Heat-pump electricity (MWh)	Net CO_2 in Finland (0.05)	Net CO_2 in EU avg (0.25)	Net CO_2 in Germany (0.36)
50%	6,067	1,103	+414 tCO_2	−2,718 tCO_2	−4,545 tCO_2
70%	8,494	1,544	+877 tCO_2	−2,488 tCO_2	−4,218 tCO_2
85%	10,314	1,875	+1,224 tCO_2	−2,316 tCO_2	−3,973 tCO_2
95%	11,528	2,096	+1,456 tCO_2	−2,200 tCO_2	−3,810 tCO_2

The cold-plate parallel reproduces the immersion conclusion: net climate benefit depends critically on grid intensity. In Finland, a 1.7 MW direct-to-chip facility with heat-pump-coupled district-heating integration delivers roughly +900 to +1,500 tCO_2/yr net displacement at base-to-ideal utilisation; in Germany the same deployment is net-negative by approximately 4,000 tCO_2/yr because the electricity to run both the AI workload and the heat pump exceeds the carbon value of the displaced gas heat. The materials-thermal argument and the grid-emissions argument both have to be in scope simultaneously for a defensible sustainability conclusion.

4.2.4. Embodied emissions of the cooling fluid

The lifecycle accounting must also include the embodied CO_2 of the cooling fluid itself. Synthetic esters and engineered hydrocarbon dielectric fluids carry an embodied burden of approximately 2–4 kg CO_2 per kg fluid based on cradle-to-gate life-cycle inventories of analogous ester and hydrocarbon products [61]. A 1.7 MW container with a fluid charge of approximately 8,000–12,000 L (density 0.8–0.9 kg L^{-1}) carries a fluid mass of 6,400–10,800 kg, corresponding to embodied emissions of 13–43 tonnes CO_2. Replacement at 5–8 year intervals adds the same embodied burden each cycle. Both figures are small relative to the 1,400–3,200 tCO_2 per year displacement base case (in a low-carbon grid), recovered within the first 1–3 weeks of operation. For two-phase fluorocarbon fluids the embodied burden per kg is several times higher and global-warming-potential leak emissions can be substantial; HFO and HFE alternatives generally have lower per-kg embodied emissions but require recharge after leak events.

4.3. Scalability and geographic constraints

Finland, Sweden, Denmark, and Norway deliver more than 50 percent of heating through district networks and have grid intensities consistent with the Section 4.2.2 breakeven analysis. Compute heat integration works in these regions because of cold climate (8–10 months annual demand), existing pipe infrastructure, supportive municipal policy, low-carbon electricity, and high conventional energy costs [4]. Many Finnish district-heating systems already source heat from co-generation plants; in these systems, the marginal climate benefit of mining heat is reduced [37]. The strongest case exists where networks depend on standalone fossil boilers, which is why the Mäntsälä Yandex installation [38] reports outsized gas-utility CO_2 reductions (40 percent), and why the Finnish district-heating sector has been an early-adopter market for this class of integration.

Germany's EnEfG implementation creates a large near-term market: approximately 1,000 data centres above the 300 kW threshold are subject to the 10/15/20 percent heat-reuse trajectory beginning in 2026 [41, 42]. As the breakeven analysis (Section 4.2.2) shows, however, the Germany grid intensity (~0.36 tCO_2/MWh) places most plausible mining-as-heating cases on the wrong side of net-zero CO_2; the EnEfG's climate logic relies on parallel grid decarbonisation continuing on schedule. The Frankfurt FRANKY case demonstrates that even a 2 percent utilisation of waste heat from a single colocation campus can supply more than 60 percent of a 3,000-resident neighbourhood's heating demand [55].

District-heating networks are designed for 30-year lifespans. Mining hardware is obsolete in 3–5 years; data-centre hardware in 7–10. Municipalities evaluating mining-heated districts must plan for multiple hardware refresh cycles per infrastructure lifecycle [18]. Bitcoin mining provides rapid demand response on a scale that other large-scale compute workloads do not match (treated in detail in Supplementary Material Section S5).

4.4. Asian deployments and disclosure asymmetry

China's data-centre liquid-cooling market reached USD 2.37 billion in 2024 (67 percent year-over-year growth) [43]. Alibaba's single-phase immersion-cooling deployments at scale have been documented to reduce energy consumption by over 35 percent compared to traditional air-cooled data centres [62]. Tencent's Tianjin facility uses a magnetic-suspension heat pump to recover waste heat for residential heating [60]. National policy under MIIT's Three-Year Action Plan for New Data Center Development (2021–2023) and successor programmes mandates a PUE under 1.3 for newly built large-scale data centres, with a national goal of 1.25 by 2025 [63]. Chinese district heating, however, remains 85 percent powered by coal and natural gas [64]; the China grid intensity (~0.55 tCO_2/MWh) places these deployments well outside the Section 4.2.2 breakeven envelope under any plausible displaced-fuel assumption, until grid decarbonisation progresses.

Disclosure asymmetry between Chinese and European data-centre operators makes deployment-level case-by-case validation difficult for the present review. Trade-press reports indicate operational immersion-cooled installations at multiple Alibaba and Tencent sites, but technical disclosure typically does not include the per-site thermal-output and output-temperature data required to populate Tables 3a/3b. Future reviews on this topic will benefit from improved disclosure norms in the Asian market.

5. Second-Generation Materials: Gallium-Based Liquid Metal Systems

5.1. Thermal performance

Pure gallium exhibits a thermal conductivity of 33.7 W m^{-1} K^{-1} in the solid state and approximately 24 W m^{-1} K^{-1} as a liquid, with a latent heat of 80.3 kJ kg^{-1} at a melting point of 29.8 °C [19]. GaInSn eutectic alloys (Galinstan®, registered to Geratherm Medical AG: 68.5 percent Ga, 21.5 percent In, 10 percent Sn) reported as thermal interface materials span 16–39 W m^{-1} K^{-1} depending on composition, bond-line thickness, surface roughness, contact pressure, and measurement protocol [20, 21]. Values toward the upper end of this range require optimised bond-line and pressure conditions and should not be taken as a generic material constant.

It is essential to distinguish two applications of gallium-based materials with very different engineering requirements and readiness levels. Ga-based thermal interface materials (TIMs), applied at the die-to-heat-sink junction in milligram quantities, are commercially mature: Sony's PlayStation 5 uses a Galinstan-based TIM in mass production, and Indium Corporation's GalliTHERM product line targets HPC semiconductor applications (TRL 8–9). Ga-based cooling loops, circulating kilograms of liquid metal through metres of facility piping, are essentially undemonstrated at scale (TRL 3–4). Bridging this gap is the central scaling challenge for the second generation. The TRL trajectory is summarised in Table 7.

Table 7. Gallium-cooling TRL trajectory.

Application	Current TRL	Representative example	Key remaining barrier	Est. TRL by 2030
TIM at die-to-heatsink interface	8–9	Sony PS5 (Galinstan); Indium GalliTHERM	Cost reduction; long-term reliability under CTE cycling	9 (mass production extends)
Lab-scale cooling loop (single-rack)	5–6	Beijing Institute of Technology rack prototypes (2023)	Containment redundancy; pump reliability	6–7
Pilot-scale cooling loop (multi-rack)	3–4	DARPA-funded prototypes	Material compatibility; leak detection; supply chain	5–6
Facility-scale cooling loop (>1 MW)	2–3	None demonstrated	All of the above plus economics	4–5

5.2. Phase-change properties and figures of merit

Boteler et al. demonstrated that for a temperature rise of 20 °C, gallium absorbs seven times more energy per unit volume than copper and three times more than organic paraffin PCMs [19]. Shao et al. developed a figure-of-merit for PCM cooling capacity; metallic alloys achieve FOMs one to two orders of magnitude higher than organic PCMs ($2{,}603 \times 10^3\ W^2\ s\ K^{-1}\ m^{-4}$ for a Bi-based metallic alloy versus 22×10^3 for paraffin wax) [22].

5.3. Experimental validation under pulsed thermal loads

Gonzalez-Nino et al. demonstrated that metallic PCMs reduced temperature rise by 60–68 percent under 19 ms pulsed thermal loads at heat fluxes up to 338 W cm^{-2}, compared to only 14–17 percent for the best organic PCM [23]. At 160 W pulse power, the metallic PCM maintained temperature within 5 °C of its melting point while the dielectric baseline rose 119 °C. The Gonzalez-Nino experiments tested 19 ms transient pulses, characteristic of switching transients in power electronics. Steady-state thermal loads at facility scale require additional analysis of melting-front propagation, density-driven convection, and supercooling-induced solidification anomalies (Section 6.4); the 60–68 percent reduction is therefore an upper bound on the steady-state benefit. Translation of pulsed-load performance to AI training workloads with multi-second to minute-scale thermal envelopes is an active research area.

Ma and Liu demonstrated thermal-conductivity enhancement ratios up to 2.3× with 20 percent CNT volume fraction in liquid gallium [24]. CNT dispersion in liquid gallium faces sedimentation, agglomeration, and oxide-entrapment challenges at volumes beyond the laboratory. The liquid-metal TIM market reached USD 305.9 million in 2025, projected to USD 479.7 million by 2032 [25]. Boteler et al.'s time-scale-matched PCM systems, pairing metallic PCMs with organic PCMs, demonstrated 25–50 percent mass reduction relative to pure metallic systems [19].

6. Engineering Challenges for Gallium-Based Cooling Systems

6.1. Corrosion and material compatibility

Gallium alloys aggressively attack aluminium via liquid-metal embrittlement. Deng and Liu found that 6063 aluminium alloy suffered complete gallium infiltration within five hours at 120 °C [29]. T2 copper exhibited slow corrosion at approximately 30–60 µm per month at 60 °C. 1Cr18Ni9 stainless steel showed zero detectable gallium infiltration after 30 days at 60 °C [29]. The practical materials palette for facility-scale cooling loops is therefore nickel-plated copper or 316L stainless steel throughout.

6.2. Electrical conductivity and containment

Liquid metals are electrically conductive ($\sim 3.5 \times 10^6$ S m^{-1}). A dielectric-fluid leak is a maintenance event; a liquid-metal leak is a catastrophic short-circuit risk. Gallium-based systems require sealed, isolated cooling channels rather than open-bath immersion. This constraint is structural: it means that gallium loops cannot adopt the open-tank architecture of first-generation immersion cooling, and must instead resemble closed-loop cold-plate systems with much tighter leak-detection requirements.

6.3. Surface tension and wettability

Gallium exhibits high surface tension (~709 mN m^{-1} vs. ~73 mN m^{-1} for water), requiring oxide-layer engineering for uniform thermal contact. Native oxide formation on gallium and Galinstan is rapid (~2 nm Ga_2O_3 within seconds of air exposure) and improves wetting on metal substrates by reducing the effective interfacial energy, but the oxide layer also increases interfacial thermal resistance.

6.4. Supercooling mitigation

Bulk gallium can supercool tens of degrees below its 29.8 °C melting point in the absence of nucleation sites. The exact extent depends strongly on purity, surface contact, and presence of nucleation sites; controlled high-purity bulk experiments have reported persistent liquid behaviour to approximately −19 °C, with deeper undercooling reported under more controlled conditions (see Zhang et al. [30] and references therein). Zhang et al. also demonstrated that 0.5 wt.% TeO_2 particles reduced supercooling by 44 percent [30], offering a practical route to a more reliable phase-change response. Supercooling is a critical concern for facility-scale loops because uncontrolled solidification could cause local blockages.

6.5. Toxicity considerations

Gallium itself exhibits low acute toxicity. Galinstan contains approximately 21.5 percent indium, and indium compounds have documented pulmonary toxicity associated with occupational exposure. The American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Value for indium and its compounds is 0.1 mg m^{-3} as an 8-hour time-weighted average [65]. ITO-specific guidance is materially tighter where in force: Japan's Ministry of Health, Labour and Welfare technical guideline (2010) recommends an indium target airborne concentration of 0.0003 mg m^{-3} for ITO production environments, motivated by the indium-lung-disease (pulmonary alveolar proteinosis) cluster documented in indium-tin-oxide manufacturing [66]. Operators planning facility-scale gallium-cooling deployments should design ventilation, monitoring, and PPE programmes to the most stringent applicable standard. At facility scale, handling protocols must address inhalation risks from aerosolised particles during alloy preparation, leak cleanup, decommissioning, and recycling, requiring local exhaust ventilation and respiratory protection per OSHA 1910.134 (or EU equivalent).

6.6. Implications for heat-reuse systems and regulatory exposure

A gallium cooling loop could capture waste heat at 80–95 °C with lower thermal resistance than first-generation systems, eliminating heat-pump boosting in some district-heating contexts and providing the modest exergy gain documented in Section 2 (Carnot fraction ~21 percent at 90 °C versus ~17 percent at 70 °C immersion). This is a projection, not yet demonstrated. Three regulatory frameworks together govern facility-scale deployment economics for gallium cooling: ECHA REACH (PFAS tightening; gallium currently unaddressed); ACGIH/OSHA/EU OEL frameworks for indium occupational exposure; and the

People's Republic of China's 1 August 2023 export controls on gallium and germanium [32], which raised gallium prices internationally and tightened supply for non-Chinese manufacturers. Operators must integrate compliance design across all three frameworks from project initiation.

7. Materials Supply and Circularity

7.1. Gallium supply chain

Gallium is recovered as a by-product of aluminium and zinc refining [26]. Historically the dominant end use has been GaAs wafer fabrication, but the USGS 2024 Mineral Commodity Summaries show GaN power electronics, RF integrated circuits, and emerging Ga_2O_3 applications now account for a growing share of demand alongside GaAs [27]. China accounts for approximately 80 percent of primary production and imposed export controls in July 2023 [27, 32]. A lifecycle analysis of global gallium flows (2000–2021) found that of 5,862 tonnes entering the industrial cycle, only 967 tonnes remain in active use, with 83.5 percent wasted and 4,446 tonnes in recoverable manufacturing or electronic waste [31]. Facility-scale cooling loops are circulatory systems in which gallium recirculates indefinitely; the operative supply-chain metric is make-up volume for leak and maintenance losses, not total fill volume, which substantially reduces ongoing demand relative to the semiconductor pathway.

7.2. Dielectric fluid lifecycle and recycling

Synthetic esters and engineered hydrocarbons used as immersion fluids are recyclable through established lubricant-reprocessing channels, with reclaim yields of 70–85 percent typical for closed-loop industrial fluids. End-of-life fluid that has accumulated TAN above 0.5 mg KOH per gram is typically incinerated for energy recovery rather than reclaimed. Two-phase fluorocarbon fluids face a different end-of-life challenge: PFAS-classified compounds cannot be incinerated below 1,100 °C without releasing PFAS-class atmospheric pollutants, and dedicated PFAS-destruction infrastructure is limited.

7.3. ASIC and GPU end-of-life

ASIC mining hardware is obsolete in 3–5 years; AI GPUs in 5–8 years; general-purpose data-centre hardware in 7–10 years [18]. de Vries and Stoll (2021) estimated approximately 30,700 tonnes of ASIC mining e-waste generated globally in 2020, with the figure growing roughly in line with hash-rate growth [67]. Recycling pathways exist for the printed circuit boards (gold, copper, palladium recovery) and for the heat-sinks and chassis, but not for the silicon dies themselves, which are typically landfilled or downcycled. Cooling-system metals (copper piping, stainless-steel manifolds, nickel plating) are generally recovered at end of life as part of the broader scrap-metal stream. Across all three generations, circularity for the cooling material itself ranges from good (single-phase dielectric fluids; gallium loops indefinitely recirculating) to challenging (two-phase fluorocarbon fluids, PFAS-restricted destruction). The dominant lifecycle burden remains the compute hardware, not the cooling material.

8. Third-Generation Materials: Embedded Two-Phase Cooling

8.1. The ICECool programme and successors

DARPA's ICECool programme demonstrated chip-level heat-flux removal exceeding 1 kW cm^{-2}, with hot-spot cooling approaching 5 kW cm^{-2} [28]. IBM demonstrated microchannels in a production microprocessor achieving junction temperatures approximately 25 °C lower than the air-cooled baseline while reducing power by more than 10 W. R1234ze(E) can produce exit temperatures in the 60–80 °C range suitable for district heating. Microsoft's collaboration with Corintis, announced in September 2025, integrates microfluidic cooling channels directly within AI accelerator silicon and reports up to three times the heat-removal capability of conventional cold plates [49]. This represents a significant TRL advance for embedded two-phase cooling, from DARPA-demonstration scale to commercial-product scale within the past two years.

Manufacturing-scalability questions remain. Through-silicon-via integration yields for microchannel-in-die designs are reportedly above 90 percent at 7 nm and 5 nm nodes for sub-millimetre die sizes, but published yield data at production AI-accelerator die sizes is limited. ICECool reliability under thermal cycling has been demonstrated above 1,000 cycles at temperature swings of 70 °C without measurable degradation in DARPA programme reports [28]; the 100,000-cycle service-life target for facility deployment has not been tested publicly. Refrigerant-loop reliability at facility scale introduces additional challenges in fluid-distribution manifold design, in particular thermal-expansion management across long runs and pressure-drop balancing across heterogeneous workloads.

8.2. Comparative materials in the design space

Several materials sit alongside the three primary generations in the materials design space (Figure 5). CVD diamond achieves thermal conductivities of 1,000–2,000 W m^{-1} K^{-1} (electronic-grade material typically 1,700–2,000), making it the highest-conductivity bulk material currently producible at engineering scale. CVD diamond and gallium liquid metals are complementary, not competitive: diamond addresses the chip-to-package interface; gallium addresses the package-to-loop and loop-to-facility transport. Hexagonal boron nitride (hBN) offers theoretical parity with diamond plus electrical insulation, but practical composites achieve only 5–15 W m^{-1} K^{-1} owing to interface scattering and orientation distribution. Graphene nanofluids achieve thermal conductivities of approximately 1–2 W m^{-1} K^{-1}, a meaningful improvement over dielectric fluids but 15–30 times below gallium. Together with the three primary generations, these alternative materials trace the materials-design space mapped in Figure 5.

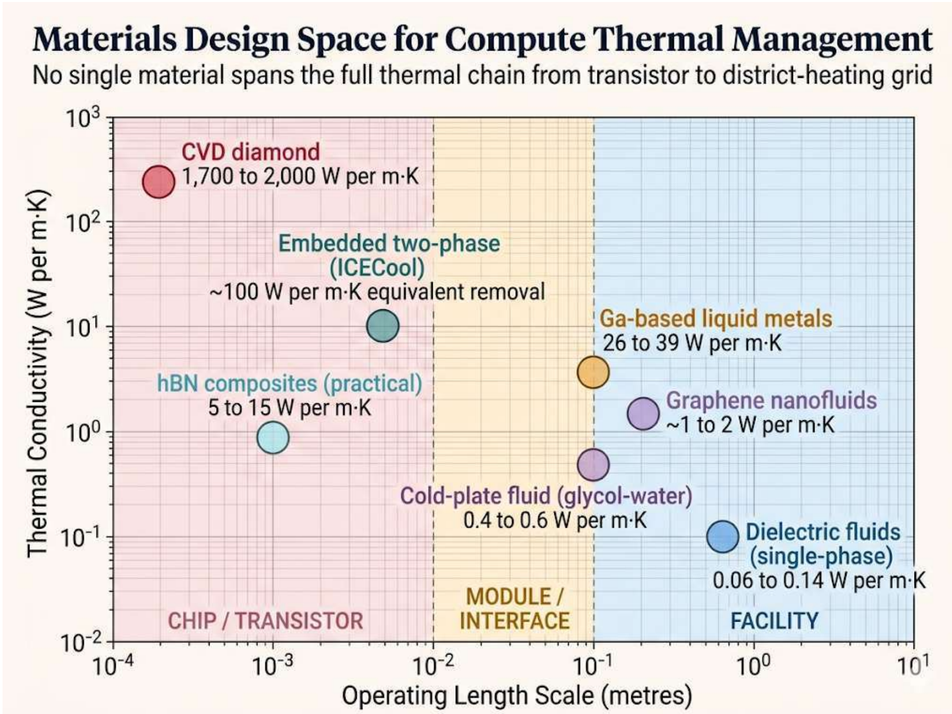


Figure 5. Materials design space for compute thermal management.

Thermal conductivity (W m⁻¹ K⁻¹) plotted against operating length scale (metres). CVD diamond at the chip and transistor scale (1,700–2,000 W m⁻¹ K⁻¹); embedded two-phase ICECool systems at the chip-to-module scale; gallium-based liquid metals at the module-to-loop scale (16–39 W m⁻¹ K⁻¹); hBN composites at the chip-to-module scale (5–15 W m⁻¹ K⁻¹ practical); graphene nanofluids at the loop scale (1–2 W m⁻¹ K⁻¹); dielectric fluids at the facility scale (0.06–0.14 W m⁻¹ K⁻¹). No single material spans the full thermal chain from transistor to district-heating grid.

9. Conclusion

Three regulatory and market shifts now make the materials-design framing developed in this review consequential beyond the laboratory. The European Union's Energy Efficiency Directive (2023/1791) mandates waste-heat recovery from new data centres above 1 MW, and Germany's EnEfG sets a binding 10/15/20 percent heat-reuse trajectory for 2026–2028. Total data-centre electricity demand, dominated by AI workloads, is projected by the IEA to approach or exceed 1,000 TWh by 2026. Within these constraints, the cooling material is the design variable that determines whether compute waste heat becomes a usable energy product or a thermodynamic loss.

First-generation materials (dielectric immersion fluids and direct-to-chip cold plates) enable the temperature-gradient breakthrough: raising waste-heat output from 25–35 °C in air-cooled facilities to 50–75 °C in liquid-cooled facilities. The Carnot exergy fraction of the recovered heat roughly doubles in the process, from ~7 percent to ~14–17 percent (Figure 2, Table 2). Seventeen operational deployments across nine countries validate this generation. The lifecycle CO_2 analyses in Sections 4.2.1–4.2.3 yield two complementary results: (i) the immersion mining case displaces approximately 1,400–3,200 tCO_2/yr per 1.7 MW container at base-case utilisation, and (ii) the parallel cold-plate AI case displaces 900–1,500 tCO_2/yr per 1.7 MW facility at the same utilisation. Both are net-positive only on grids cleaner than approximately 0.13–0.24 tCO_2/MWh — a condition met in the Nordic region and France but not in Germany, the United States, or China at present (Section 4.2.2 breakeven matrix).

Second-generation gallium-based liquid metals (16–39 W m^{-1} K^{-1} at the thermal interface) offer roughly two orders of magnitude higher conductivity than dielectric fluids, with 60–68 percent temperature-rise reduction under pulsed loads. Mature TIM applications (TRL 8–9) must be clearly distinguished from undemonstrated facility-scale cooling loops (TRL 3–4). Engineering challenges in corrosion, electrical conductivity, indium occupational toxicity, and supply-chain concentration are tractable but require integrated design-to-deployment compliance. Third-generation embedded two-phase microfluidic cooling, demonstrated at over 1 kW cm^{-2} under DARPA's ICECool programme and now reaching commercial scale through the 2025 Microsoft–Corintis collaboration, points toward chip-level thermal management at heat-flux densities that facility-level immersion cannot reach.

Four open research and policy questions will shape the next decade of work in this area. (1) Can gallium-based facility-scale cooling loops be demonstrated at TRL 6–7 within a five-year horizon, or do the engineering challenges (containment, electrical safety, supply chain) lock gallium into the TIM-and-prototype niche? (2) Can embedded two-phase microfluidic cooling reach manufacturing yields and reliability metrics that justify integration into production AI accelerators at hyperscale, beyond the Microsoft–Corintis pilot? (3) Can district-heating tariff structures evolve to compensate compute operators for delivered heat at scale, beyond the open-tariff model pioneered by Stockholm Exergi? (4) Can deployment-level CO_2 verification become a standard practice, so that vendor projections such as the Microsoft–Fortum 400,000 tCO_2/yr figure can be independently audited the way the AWS Dublin scheme has been — closing the gap between the green entries of Table 3a and the gray entries of Table 3b, and putting the breakeven calculation of Section 4.2.2 on measured rather than modelled footing?

The materials choice is not merely a thermal-engineering decision but a sustainable energy-systems design decision, with downstream consequences for water consumption, carbon displacement, and district-heating compatibility, and whose net climate value is bounded by both Carnot exergy on the heat side and grid-emissions intensity on the electricity side. The materials-design frontier documented here is the engineering path along which compute infrastructure will continue to shed its thermal waste profile and integrate into the broader sustainable energy system.

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The author holds no material positions in Bitcoin, mining companies, or energy firms discussed herein.

Data Sources

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Use of AI-Assisted Tools

The author used Anthropic Claude (Opus and Sonnet variants) for editorial revision (sentence-level rewriting for clarity and concision), structural reorganisation (section sequencing, heading hierarchy, table formatting), figure-production assistance (Python plot scripting, including the Carnot overlay in Figure 2 and the PRISMA flow diagram in Figure S1; figure-caption drafting), and manuscript formatting. All factual claims, numerical values, citations, the three-generation taxonomy, the lifecycle CO_2 and breakeven models, the parallel cold-plate case study, the Carnot exergy analysis, the materials-property comparisons, and the analytical conclusions are the author's work. No AI tool generated factual claims, citations, or numerical values that were not subsequently verified by the author against the primary sources. Disclosure language follows RSER's 2025 guidance on generative AI use in submitted manuscripts.

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Supplementary Material

Supplementary_Material_REVISED.docx accompanies this manuscript and contains: S1 PRISMA-compatible flow diagram (Figure S1); S2 Sensitivity of Carnot calculations to T_0; S3 Breakeven derivation for Section 4.2.2; S4 Cold-plate parallel case-study assumptions for Section 4.2.3; S5 Demand-response and grid-integration analysis (ERCOT large-load curtailment data, Riot Platforms 700 MW August 2023 curtailment, USD 31.7M earned, 2025 ERCOT large-load applications totalling 226 GW with 73 percent AI).